Lecture 12 Scribe

L12_S1_A.pdf
PDF

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Instructions:

1. Purpose: The scribe must be exam-ready, strictly following the content of the slides.

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3. Constraints: Do not introduce new examples, outside information, or hallucinations. Do not use phrases like 'for example' to add your own content.

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Goal: Produce a clean, accurate revision document based solely on the attached file."

Below is an **exam-ready lecture scribe** prepared strictly from the provided Lecture 12 slides.

CSE400 – Fundamentals of Probability in Computing

Lecture L11–12: Transformation of Random Variables L12_S1_A

1. Transformation of Random Variables

Goal

Given a random variable X with known PDF $f_X(x)$ and CDF $F_X(x)$, and a transformation

$$Y = g(X)$$

find the distribution of Y: $F_Y(y)$ and $f_Y(y)$. L12_S1_A

Case: Monotonic Transformation

Assume $g(x)$ is invertible.

Step 1: CDF Method

$$F_Y(y) = P(Y \leq y)$$

Since $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

If g is increasing:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

If g is decreasing:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Step 2: PDF via Differentiation

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

For invertible g:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

where

$$x = g^{-1}(y)$$

Notes from slides

- Limits must be transformed appropriately.
- Sign of derivative handled using absolute value.
- Applies when $f_X(x)$ is known a priori. L12_S1_A

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2. Worked Example: Nonlinear Transformation

Given

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

L12_S1_A

Step 1: Inverse Transformation

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 2: Derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3: Apply Transformation Formula

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

$$= \frac{1}{2}\cdot\frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

L12_S1_A

3. Function of Two Random Variables

Let

$$Z = g(X, Y)$$

Examples shown:

- $Z = X + Y$
- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

L12_S1_A

4. Distribution of the Sum: $Z = X + Y$

CDF Approach

$$F_Z(z) = P(Z \leq z)$$

$$= P(X + Y \leq z)$$

$$= \iint_{x+y \le z} f_{XY}(x, y)\, dx\, dy$$

L12_S1_A

Graphical region integration is illustrated on the slides. L12_S1_A

PDF via Differentiation and Leibniz Rule

Using differentiation under the integral:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx$$

L12_S1_A

5. Convolution Formula (Independent Case)

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

then

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z - x)\, dx}$$

This is the **convolution integral**. L12_S1_A

6. Special Case: Sum of Normal Random Variables

Given:

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

independent.

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

After completing the square and integrating:

$$\boxed{Z \sim N(0, 2)}$$

L12_S1_A

7. Sum of Exponential Random Variables

If

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

then

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$\boxed{f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z \geq 0}$$

$$f_Z(z) = 0 \text{ otherwise}$$

L12_S1_A

8. Difference of Random Variables

For

$$Z = X - Y$$

$$F_Z(z) = P(X - Y \leq z)$$

$$= \iint_{x-y \leq z} f_{XY}(x, y)\, dx\, dy$$

PDF obtained by differentiation (same procedure as sum). L12_S1_A

9. Ratio Transformation

For

$$Z = \frac{X}{Y}$$

analysis depends on the sign of Y (separate cases for $Y > 0$ and $Y < 0$). L12_S1_A

10. Radial Transformation Example

Slides present:

$$R = \sqrt{X^2 + Y^2}$$

Find $f_R(r)$. L12_S1_A

11. Leibniz Rule (Used in Derivations)

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)\, dy$$

then

$$\frac{dG(x)}{dx} = h(x, b(x))\frac{db}{dx} - h(x, a(x))\frac{da}{dx} + \int_{a(x)}^{b(x)} \frac{\partial h(x, y)}{\partial x} dy$$

L12_S1_A

Key Takeaways

- Transformations allow derivation of distributions of new random variables.
- CDF method is primary; PDF obtained by differentiation.
- For monotonic transformations: use inverse and Jacobian.
- For functions of two RVs: integrate over appropriate region.
- Sum of independent RVs → convolution.
- Normal sums remain normal; exponential sums yield Gamma-type form.

If you want, I can also:

✅ simplify this into **short revision notes**
✅ explain each derivation step-by-step
✅ give likely **exam questions from this lecture**
✅ summarize formulas on one page

Sources

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CSE400 – Fundamentals of Probability in Computing

Lecture L11–12: Transformation of Random Variables L12_S1_A

Outline L12_S1_A

1. **Transformation of Random Variables**
   Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for the case $Z = X + Y$.

1. Transformation of Random Variables

Problem Setup

Given a random variable X with known PDF $f_X(x)$ and CDF $F_X(x)$, and a transformation

$$Y = g(X)$$

determine the distribution of Y: $F_Y(y)$ and $f_Y(y)$. L12_S1_A

Step 1: CDF Method

$$F_Y(y) = P(Y \leq y)$$

Since $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

If $g(x)$ is monotonically increasing

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

If $g(x)$ is monotonically decreasing

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Step 2: PDF of Transformed Variable

Differentiate the CDF:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

For invertible transformations:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|}$$

where $x = g^{-1}(y)$. L12_S1_A

Notes from Slides

- Adjust limits after transformation.
- Use absolute value of derivative.
- Applicable when PDF of X is known a priori. L12_S1_A

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2. Worked Example: Nonlinear Transformation

Given L12_S1_A

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Inverse Transformation

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

PDF of Y

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

$$= \frac{1}{2}\cdot\frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

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3. Function of Two Random Variables

Consider a new variable

$$Z = g(X, Y)$$

Transformations illustrated in slides include: L12_S1_A

- $Z = X + Y$
- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

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4. Distribution of the Sum $Z = X + Y$

CDF Method L12_S1_A

$$F_Z(z) = P(Z \le z)$$

$$= P(X + Y \le z)$$

$$= \iint_{x+y\le z} f_{XY}(x, y)\, dx\, dy$$

The slides depict the integration region graphically (page 7).

Using Leibniz Rule to Differentiate

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)\, dy$$

then

$$\frac{dG}{dx} = h(x, b(x))\frac{db}{dx} - h(x, a(x))\frac{da}{dx} + \int_{a(x)}^{b(x)} \frac{\partial h}{\partial x}\, dy$$

L12_S1_A

This rule is used to obtain the PDF from the CDF.

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5. PDF of $Z = X + Y$

After differentiation:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx}$$

L12_S1_A

Independent Case (Convolution Integral)

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is labeled the **convolution integral** in the slides.

L12_S1_A

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6. Sum of Independent Normal Random Variables

Given:

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

Using convolution and completing the square:

$$\boxed{Z \sim N(0, 2)}$$

L12_S1_A

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7. Sum of Independent Exponential Random Variables

If

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

then

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$
$$= \lambda^2 z e^{-\lambda z}, \quad z \geq 0$$
$$f_Z(z) = 0 \text{ otherwise}$$

L12_S1_A

8. Difference Transformation

For

$$Z = X - Y$$
$$F_Z(z) = P(X - Y \leq z)$$
$$= \iint_{x-y \leq z} f_{XY}(x, y)\, dx\, dy$$
$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Procedure is the same as for $Z = X + Y$. L12_S1_A

9. Ratio Transformation

For

$$Z = \frac{X}{Y}$$

CDF setup:

$$F_Z(z) = P\left(\frac{X}{Y} \leq z\right)$$

Analysis requires separate consideration of

- $Y > 0$
- $Y < 0$ L12_S1_A

10. Radial Transformation

Example shown:

$$R = \sqrt{X^2 + Y^2}$$

Find $f_R(r)$.
L12_S1_A

Key Results Summary

Transformation (single RV)

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

Sum of two RVs

$$f_Z(z) = \int f_{XY}(x, z - x)\, dx$$

Independent case (Convolution)

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Special results

- $N(0,1) + N(0,1) \to N(0,2)$
- Sum of exponentials:

$$f_Z(z) = \lambda^2 z e^{-\lambda z},\ z \geq 0$$

This scribe follows the lecture flow and derivations exactly as presented in the slides.

Sources

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Lecture L11–12: Transformation of Random Variables L12_S1_A

Outline L12_S1_A

1. **Transformation of Random Variables** – learning transformation techniques.
2. **Function of Two Random Variables** – joint transformations and derived distributions.
3. **Illustrative Example** – detailed derivation for the case $Z = X + Y$.

1. Transformation of Random Variables

Problem Setup

Given a continuous random variable X with known PDF $f_X(x)$ and CDF $F_X(x)$, define

$$Y = g(X)$$

The objective is to determine $F_Y(y)$ and $f_Y(y)$. L12_S1_A

CDF Method

$$F_Y(y) = P(Y \leq y)$$

Since $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

If $g(x)$ is monotonic increasing

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

If $g(x)$ is monotonic decreasing

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

PDF of Transformed Variable

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

For invertible transformations:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|}$$

with $x = g^{-1}(y)$. L12_S1_A

Notes

- Limits must be changed appropriately.
- Absolute value ensures non-negative density.
- Applicable when $f_X(x)$ is known.

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2. Worked Transformation Example

Given L12_S1_A

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Inverse Mapping

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

Resulting PDF

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

$$= \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1 - y^2}}, \quad -1 < y < 1}$$

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3. Function of Two Random Variables

Define a new variable

$$Z = g(X, Y)$$

Transformations shown in the slides include: L12_S1_A

- $Z = X + Y$
- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

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4. Distribution of the Sum $Z = X + Y$

CDF Formulation

L12_S1_A

$$F_Z(z) = P(Z \leq z)$$

$$= P(X + Y \leq z)$$

$$= \iint_{x+y \leq z} f_{XY}(x, y)\, dx\, dy$$

The shaded integration region is illustrated graphically on page 7.

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5. Leibniz Rule (Used in Derivation)

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)\, dy$$

then

$$\frac{dG(x)}{dx} = h(x, b(x))\frac{db}{dx} - h(x, a(x))\frac{da}{dx} + \int_{a(x)}^{b(x)} \frac{\partial h(x, y)}{\partial x}\, dy$$

L12_S1_A

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6. PDF of $Z = X + Y$

Differentiating the CDF:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx}$$

L12_S1_A

Independent Case (Convolution)

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This integral is identified as the **convolution integral**.

7. Sum of Independent Normal Variables

Given:

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

Using convolution and completing the square:

$$\boxed{Z \sim N(0, 2)}$$

L12_S1_A

8. Sum of Independent Exponential Variables

If

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

then

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

$$\boxed{f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z \geq 0}$$

$$f_Z(z) = 0 \text{ otherwise}$$

L12_S1_A

9. Difference Transformation

For

$$Z = X - Y$$

$$F_Z(z) = P(X - Y \leq z)$$

$$= \iint_{x-y\leq z} f_{XY}(x, y)\, dx\, dy$$

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Procedure is identical to the sum case. L12_S1_A

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10. Ratio Transformation

For

$$Z = \frac{X}{Y}$$

$$F_Z(z) = P\left(\frac{X}{Y} \leq z\right)$$

Evaluation requires separate treatment of:

- $Y > 0$
- $Y < 0$ L12_S1_A

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11. Radial Transformation

Transformation shown:

$$R = \sqrt{X^2 + Y^2}$$

Find $f_R(r)$. L12_S1_A

Key Results

Single-variable transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left|\frac{dx}{dy}\right|$$

Sum of two variables

$$f_Z(z) = \int f_{XY}(x, z - x)\, dx$$

Independent case (convolution)

$$f_Z(z) = \int f_X(x) f_Y(z-x)\, dx$$

Derived results

$$N(0,1) + N(0,1) \to N(0,2)$$

$$f_Z(z) = \lambda^2 z e^{-\lambda z},\ z \geq 0$$

This document follows the lecture flow and reproduces the derivations and results shown in the slides for exam preparation.

Sources